



A Review of Generative AI Powered Testing Methods in Web Development

Dean Patterson



Focus of the Paper

- Generative AI is already used in software testing for tasks like creating unit tests or user stories.
- Web development has special cases generative AI testing is beneficial for
- Because generative AI for web development testing is a smaller and more recent subset of generative AI testing, it would be beneficial to compare some of the more recent methods.



Research Procedure

Paper searching process:

Search String: ("generative AI" OR "gen ai" OR "generative artificial intelligence" OR "gen artificial intelligence" OR "large language model" OR GenAI) ("web testing" OR "web test" OR "website testing" OR "website test") -"security testing" -"penetration testing"

Filtered for review articles between 2023 - 2026

Papers found: 12

After snowballing: 20

Filtering: Papers located in reputable journals, web development as the main objective of the paper, some mention of generative AI techniques

Final Number of papers: 14



Cypress Copilot (Nettur et al., 2025)

- Generate code for web testing in BDD Cypress framework
 - step definition files
 - POM files
- GPT-4o + few-shot chain prompt performed the best
- Drawbacks: Code needs to be structured to accommodate for the kind of output the OpenAI models give, adjustments still needed after code generation



End-to-end automated UI testing workflow for web sites with intensive user–system interactions (Oguz et al., 2022)

- End to end automated UI testing script generation
- User path prediction
- Drawbacks:
 - Because it is dependent on website structure, unique or complicated websites may cause the AI to struggle
 - Only three websites were tested in this paper



Evaluation of the choice of LLM in a multi-agent solution for GUI-test generation (Tomic et al., 2025)

- Multi-Agent LLM framework(PathFinder)
 - Sought to combine different local LLMs with the goal of purchasing an item from an e-commerce website
 - The framework allows for four LLMs
 - perception and summarization
 - decision-making
 - input handling and extraction
 - validation
 - Found using one LLM for all agents was better for simple websites and different LLMs across the agents was better for more complicated websites.



VON Similo LLM (Nass et al., 2023)

- Aims to improve GUI testing
- Uses an LLM to attempt to locate web elements
- -ranks candidates with VON Similo
- -Converts best to JSON
- -Create LLM prompt
- -Get best
- Greater accuracy than baseline algorithm but more expensive and time consuming

T. Li, C. Cui, R. Huang, D. Towey, L. Ma, Large language models for automated web-form-test generation: an empirical study, ACM Trans. Software Eng. Method

- web form testing
- HTML input should be structured into LLM friendly prompts
- Non-GPT LLMs should be used to prevent data leakage if including real data as part of testing



Leveraging large language models for explicit wait management in end-to-end web testing (Olianas et al., 2025)

- A way to replace "thread sleeps" with explicit waits in Selenium
 - Starts with a prompt with low context to ChatGPT asking for how to replace the thread sleeps
 - Test script is run ten times
 - If the script fails ChatGPT is asked successive times each with more context
 - If the prompts are all used, ChatGPT is given the error output of each attempt and asked to try again



Leveraging large vision-language model for better automatic web GUI testing (Wang et al., 2024)

- Uses a vision language model to understand and test the GUI
 - Target element selector
 - Text input generator
- Overcomes issues with regular testing where the types for text fields are often not well defined



Vision-Based Mobile App GUI Testing: A Survey (Yu et al. 2025)

- Outputs of LLMs can be difficult for automated systems to understand
- An LLM might offer results that are too general when more specialized output is desired
- A major hurdle is making sure that generated test cases, execution, and verification steps all align



Comparison

- Token Limits
- Prompt Engineering
- Variety of Uses
- Data Privacy Concerns



Limitations of the Study

- This study was limited to only testing methods specialized to web development
- Security testing ommited as the focus was on only website testing



Conclusion and Future Work

- Overall Benefits: can result in more complete, accurate, and fast testing
- Overall Detriments: comes with the downsides of LLMs(privacy concerns, tokens)
- Multimodal approaches